Stroke Volume

Prodigy Anesthesia

The stroke volume is the amount of blood that is ejected out of the heart during systole. At the end of diastole the ventricles are full of blood. The total amount of blood in the ventricles at the end of diastole is called the end-diastolic volume. During systole this blood is ejected out of the ventricles. The amount of blood remaining in the ventricles at the end of systole is called the end-systolic volume. The stroke volume equals the end-diastolic volume minus the end-systolic volume.

Stroke Volume = end diastolic volume – end systolic volume

Normally, the stroke volume ranges from about 60 to 130 mL/beat. The primary factors that can affect stroke volume are changes in contractility, heart rate, afterload, and preload.

Calculate the stroke volume in a patient with an end-diastolic volume of 142 mL and an end-systolic volume of 55 mL.

Stroke Volume = EDV – ESV

Stroke Volume = 142 mL – 55 mL

Stroke Volume = 87 mL

This patient has a stroke volume of 87 mL, which falls within the normal range.

Stroke Volume Calculation Practice Problems

Calculate the stroke volume when:

1. The EDV = 135 and the ESV = 42
2. The EDV = 118 and the ESV = 49
3. The EDV = 123 and the ESV = 50
4. The EDV = 142 and the ESV = 63
5. The EDV = 129 and the ESV = 54
6. The EDV = 131 and the ESV = 38
7. The EDV = 150 and the ESV = 71
8. The EDV = 138 and the ESV = 60
9. The EDV = 124 and the ESV = 41
10. The EDV = 172 and the ESV = 83

Stroke Volume Calculation Answers

1. 93 m
2. 69 mL
3. 73 mL
4. 79 mL
5. 75 mL
6. 93 mL
7. 79 mL
8. 78 mL
9. 83 mL
10. 89 mL